

STAT 4620 Final Project

Samhit Kasichainula

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Load Libraries

```
library(tidyverse)
library(janitor)
library(GGally)
library(skimr)
library(caret)
library(naniar)
```

```
load("Wage_Stat4620_2023.RData")
Wage <- Wage_Stat4620
head(Wage)
```

```
##           X year age           maritl      race      education      region
## 1 231655 2006  18 1. Never Married 1. White    1. < HS Grad 2. Middle Atlantic
## 2  86582 2004  24 1. Never Married 1. White    4. College Grad 2. Middle Atlantic
## 3 161300 2003  45      2. Married 1. White    3. Some College 2. Middle Atlantic
## 4 155159 2003  43      2. Married 3. Asian    4. College Grad 2. Middle Atlantic
## 5  11443 2005  50      4. Divorced 1. White      2. HS Grad 2. Middle Atlantic
## 6 376662 2008  54      2. Married 1. White    4. College Grad 2. Middle Atlantic
##           jobclass      health health_ins logwage      wage      Resp
## 1  1. Industrial      1. <=Good      2. No 4.318063  75.04315 28.024
## 2  2. Information  2. >=Very Good      2. No 4.255273  70.47602 29.064
## 3  1. Industrial      1. <=Good      1. Yes 4.875061 130.98218 36.118
## 4  2. Information  2. >=Very Good      1. Yes 5.041393 154.68529 38.678
## 5  2. Information      1. <=Good      1. Yes 4.318063  75.04315 29.526
## 6  2. Information  2. >=Very Good      1. Yes 4.845098 127.11574 41.816
```

```
summary(Wage)
```

```
##           X           year           age           maritl
## Min.      : 7373   Min.      :2003   Min.      :18.00   Length:3000
## 1st Qu.: 85622   1st Qu.:2004   1st Qu.:33.75   Class :character
## Median :228800   Median :2006   Median :42.00   Mode  :character
## Mean      :218883   Mean      :2006   Mean      :42.41
## 3rd Qu.:374760   3rd Qu.:2008   3rd Qu.:51.00
## Max.      :453870   Max.      :2009   Max.      :80.00
##
##           race           education           region           jobclass
## Length:3000   Length:3000           Length:3000           Length:3000
```

```
## Class :character   Class :character   Class :character   Class :character
## Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##      health          health_ins          logwage          wage
## Length:3000        Length:3000        Min.   :3.000        Min.   : 20.09
## Class :character    Class :character    1st Qu.:4.447        1st Qu.: 85.38
## Mode  :character    Mode  :character    Median :4.653        Median :104.92
##                                     Mean  :4.654        Mean  :111.70
##                                     3rd Qu.:4.857        3rd Qu.:128.68
##                                     Max.   :5.763        Max.   :318.34
##
##      Resp
## Min.   : 1.00
## 1st Qu.:30.12
## Median :36.95
## Mean   :35.66
## 3rd Qu.:39.47
## Max.   :59.11
## NA's   :60
```

```
# Count NAs per variable
```

```
Wage %>% summarise(across(everything(), ~sum(is.na(.))))
```

```
## X year age maritl race education region jobclass health health_ins logwage
## 1 0 0 0 0 0 0 0 0 0 0 0
## wage Resp
## 1 0 60
```

Convert categorical variables to factors

```
cat_vars <- c("maritl", "race", "education",
              "jobclass", "health", "health_ins")

Wage <- Wage %>%
  mutate(across(all_of(cat_vars), as.factor)) %>% select(-region)

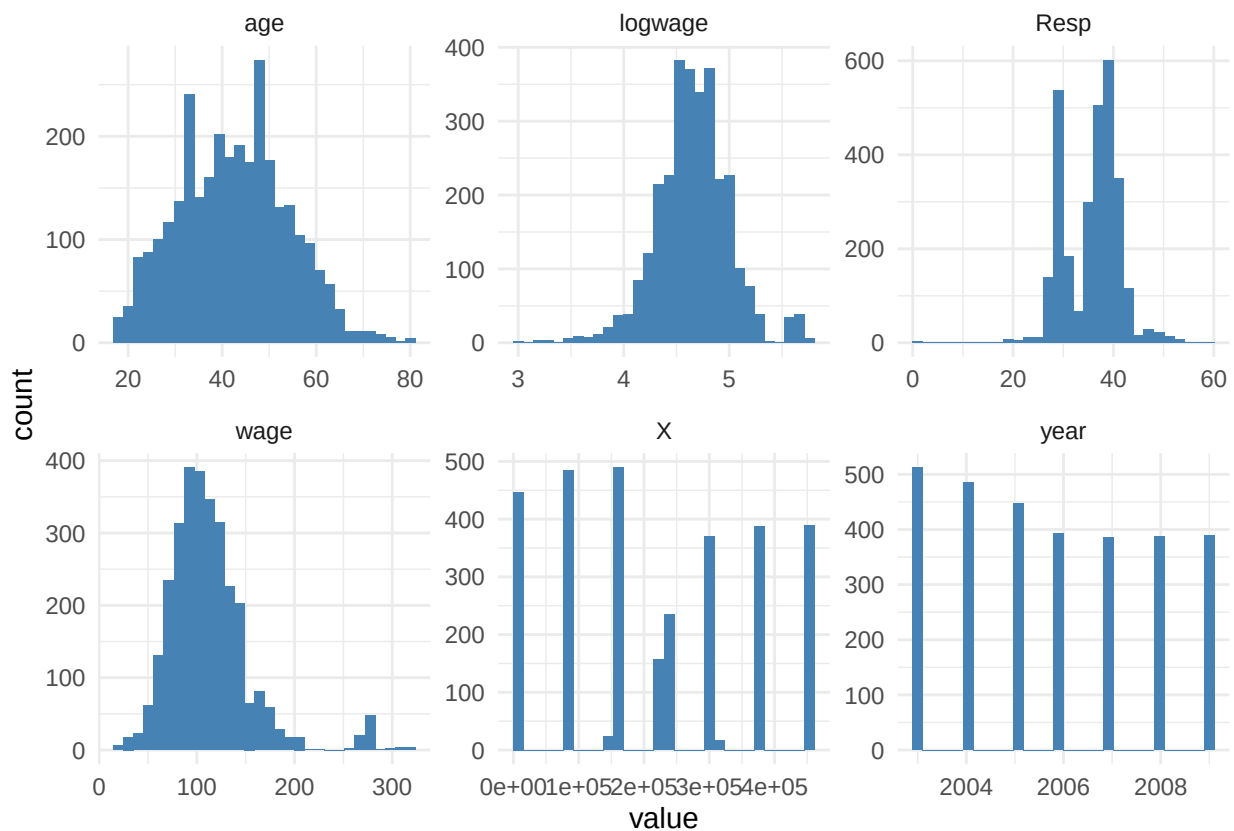
str(Wage)
```

```
## 'data.frame': 3000 obs. of 12 variables:
## $ X : int 231655 86582 161300 155159 11443 376662 450601 377954 228963 81404 ...
## $ year : int 2006 2004 2003 2003 2005 2008 2009 2008 2006 2004 ...
## $ age : int 18 24 45 43 50 54 44 30 41 52 ...
## $ maritl : Factor w/ 5 levels "1. Never Married",...: 1 1 2 2 4 2 2 1 1 2 ...
## $ race : Factor w/ 4 levels "1. White","2. Black",...: 1 1 1 3 1 1 4 3 2 1 ...
## $ education : Factor w/ 5 levels "1. < HS Grad",...: 1 4 3 4 2 4 3 3 3 2 ...
## $ jobclass : Factor w/ 2 levels "1. Industrial",...: 1 2 1 2 2 2 1 2 2 2 ...
## $ health : Factor w/ 2 levels "1. <=Good","2. >=Very Good": 1 2 1 2 1 2 2 1 2 2 ...
## $ health_ins: Factor w/ 2 levels "1. Yes","2. No": 2 2 1 1 1 1 1 1 1 1 ...
## $ logwage : num 4.32 4.26 4.88 5.04 4.32 ...
## $ wage : num 75 70.5 131 154.7 75 ...
## $ Resp : num 28 29.1 36.1 38.7 29.5 ...
```

Histograms for all numeric variables

```
Wage %>%
  select(where(is.numeric)) %>%
  pivot_longer(everything()) %>%
  ggplot(aes(value)) +
  geom_histogram(bins = 30, fill = "steelblue") +
  facet_wrap(~name, scales = "free") +
  theme_minimal()
```

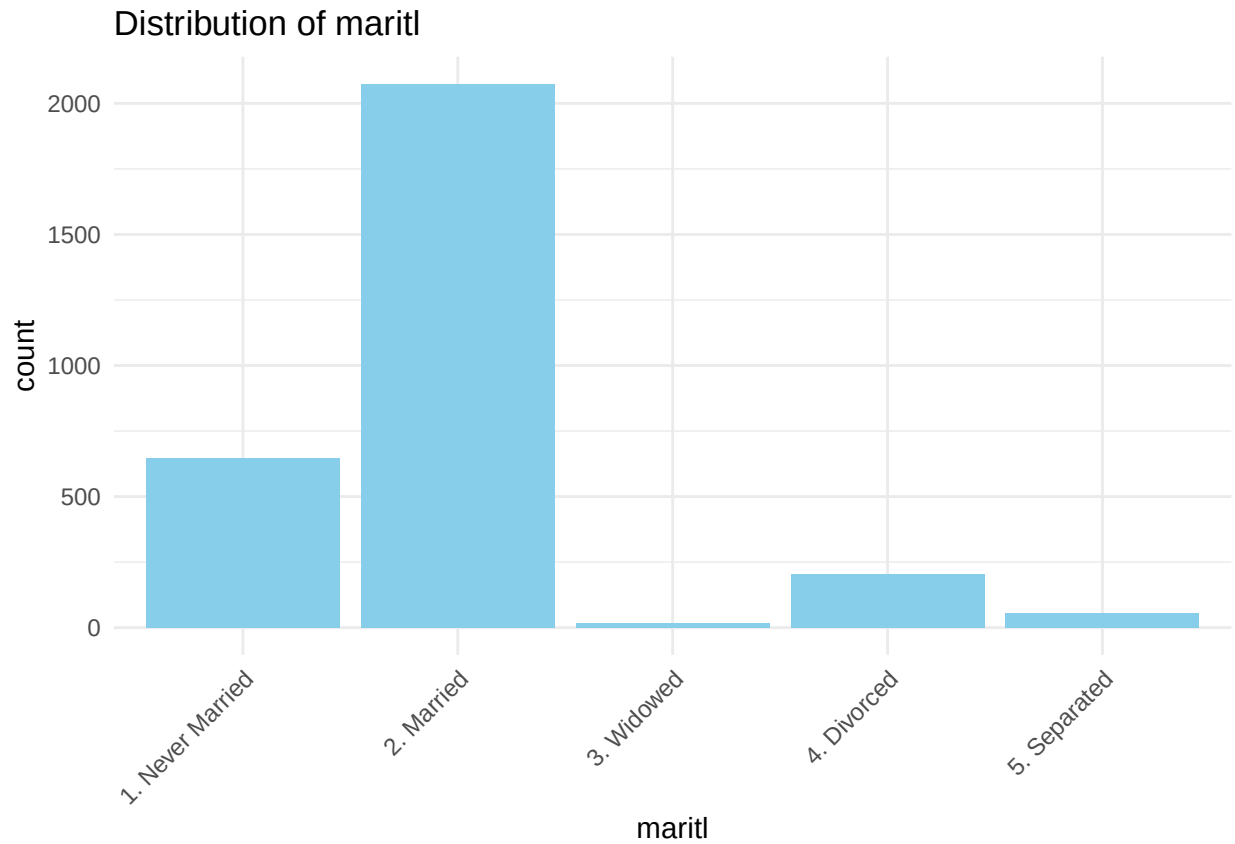
```
## Warning: Removed 60 rows containing non-finite outside the scale range
## ('stat_bin()').
```

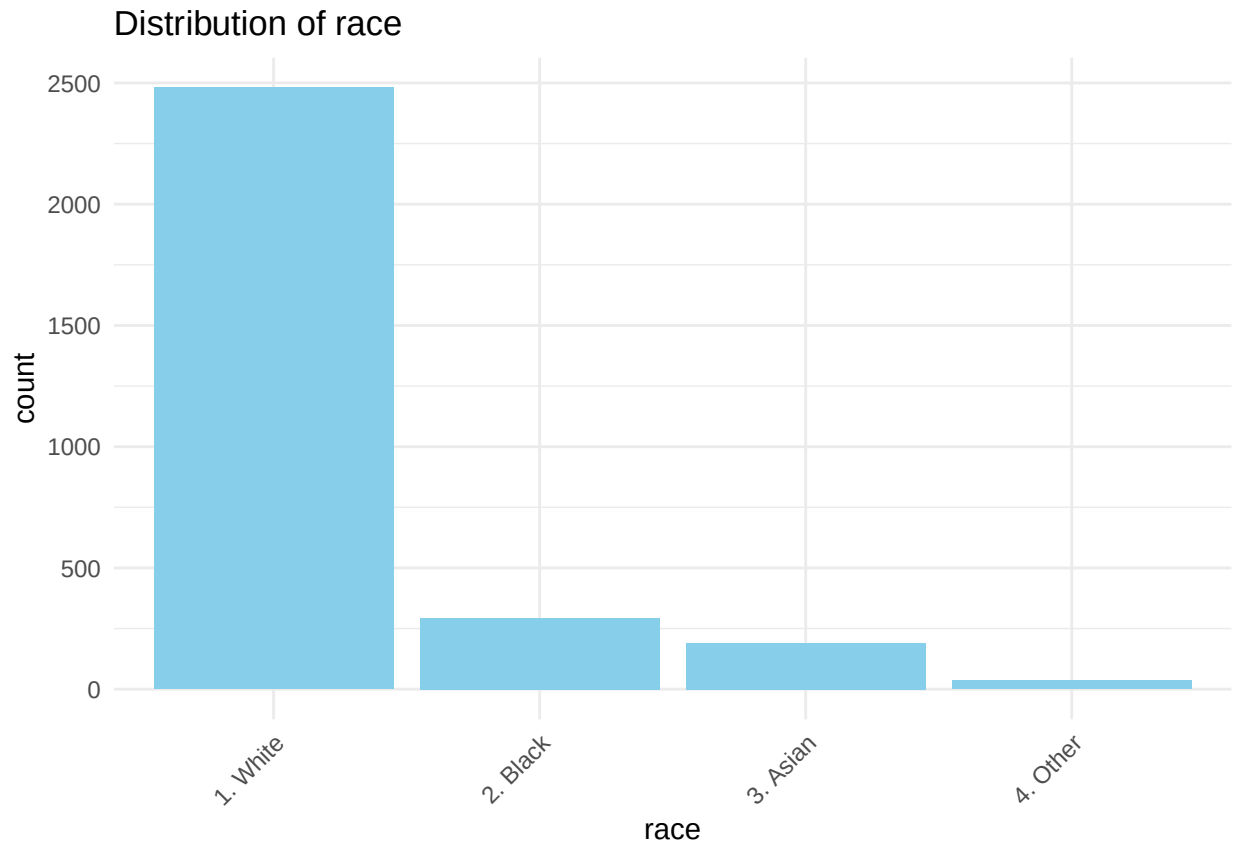


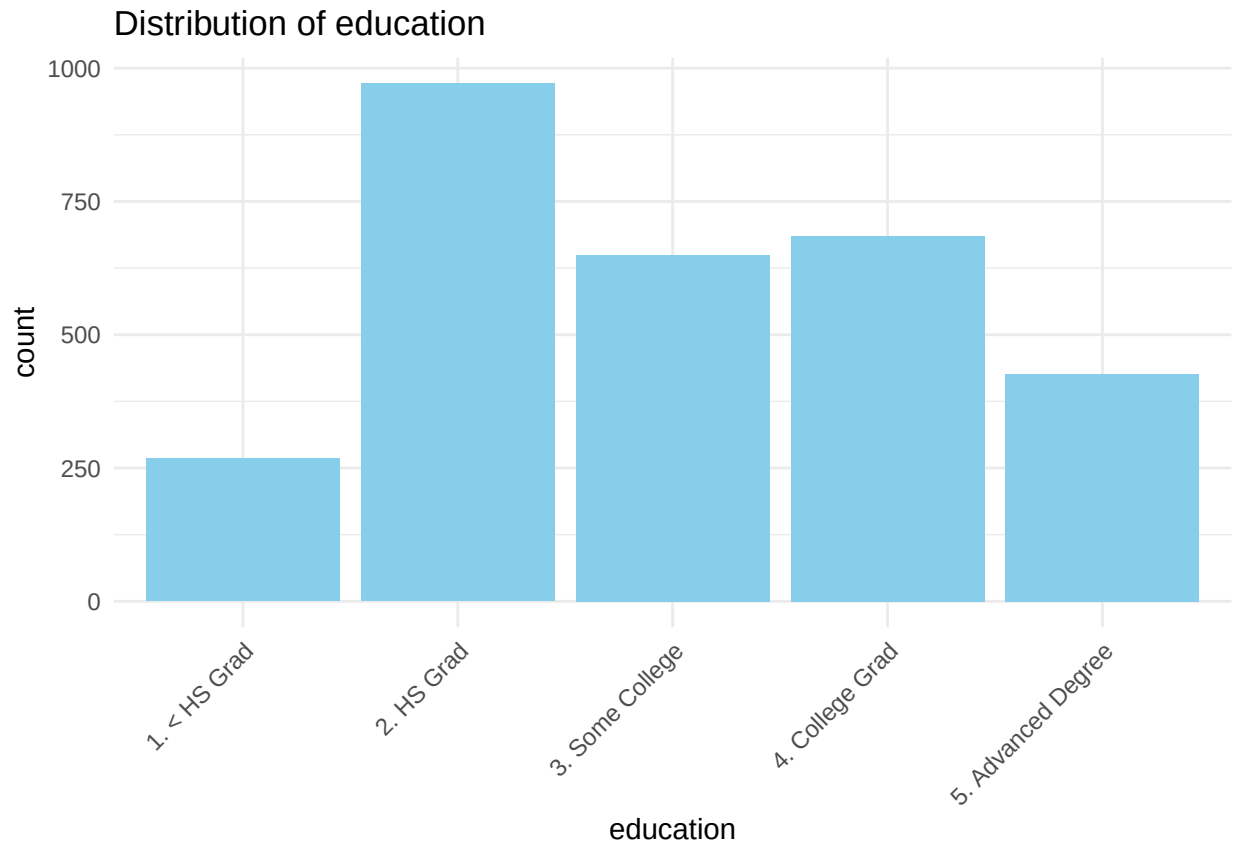
Bar plots for all categorical variables

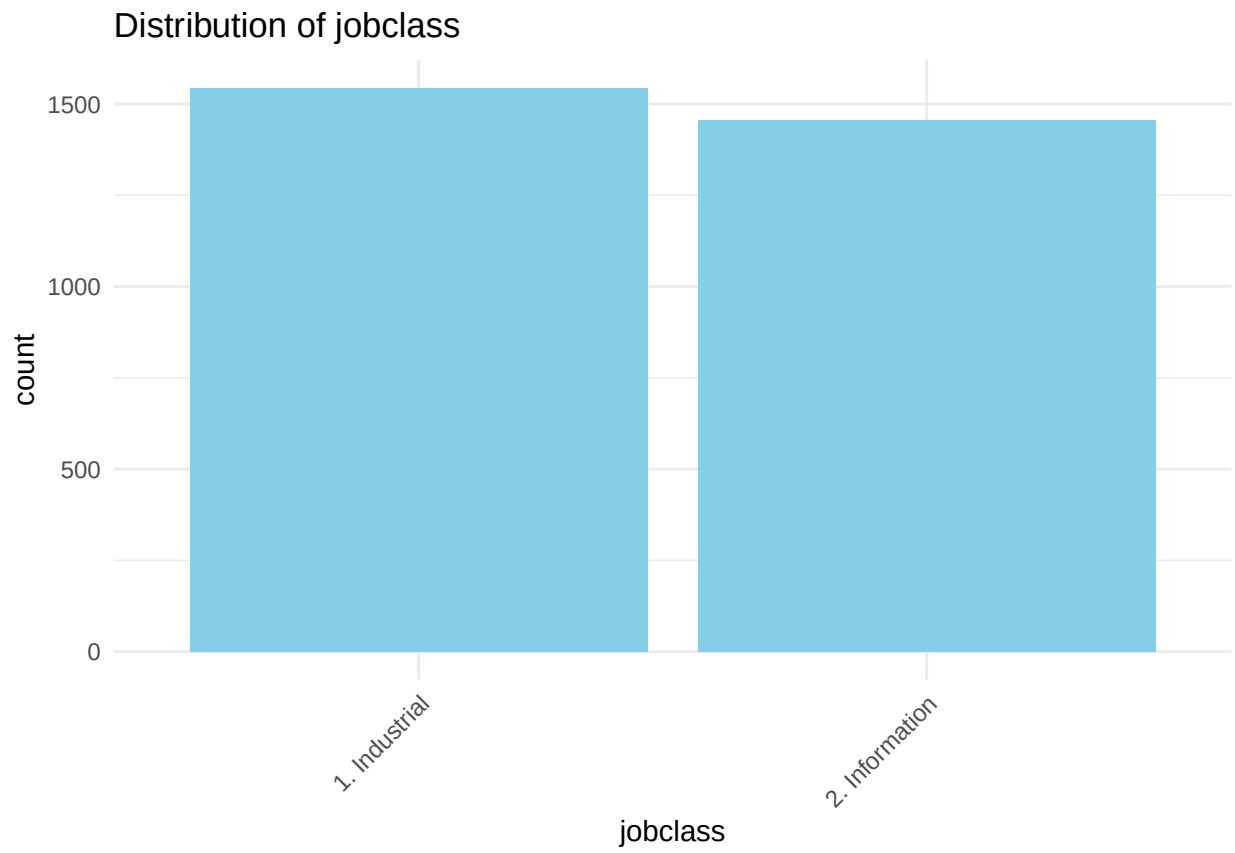
```
for (v in cat_vars) {
  print(
    Wage %>%
      ggplot(aes_string(x = v)) +
      geom_bar(fill = "skyblue") +
      theme_minimal() +
      theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
      ggtitle(paste("Distribution of", v))
  )
}
```

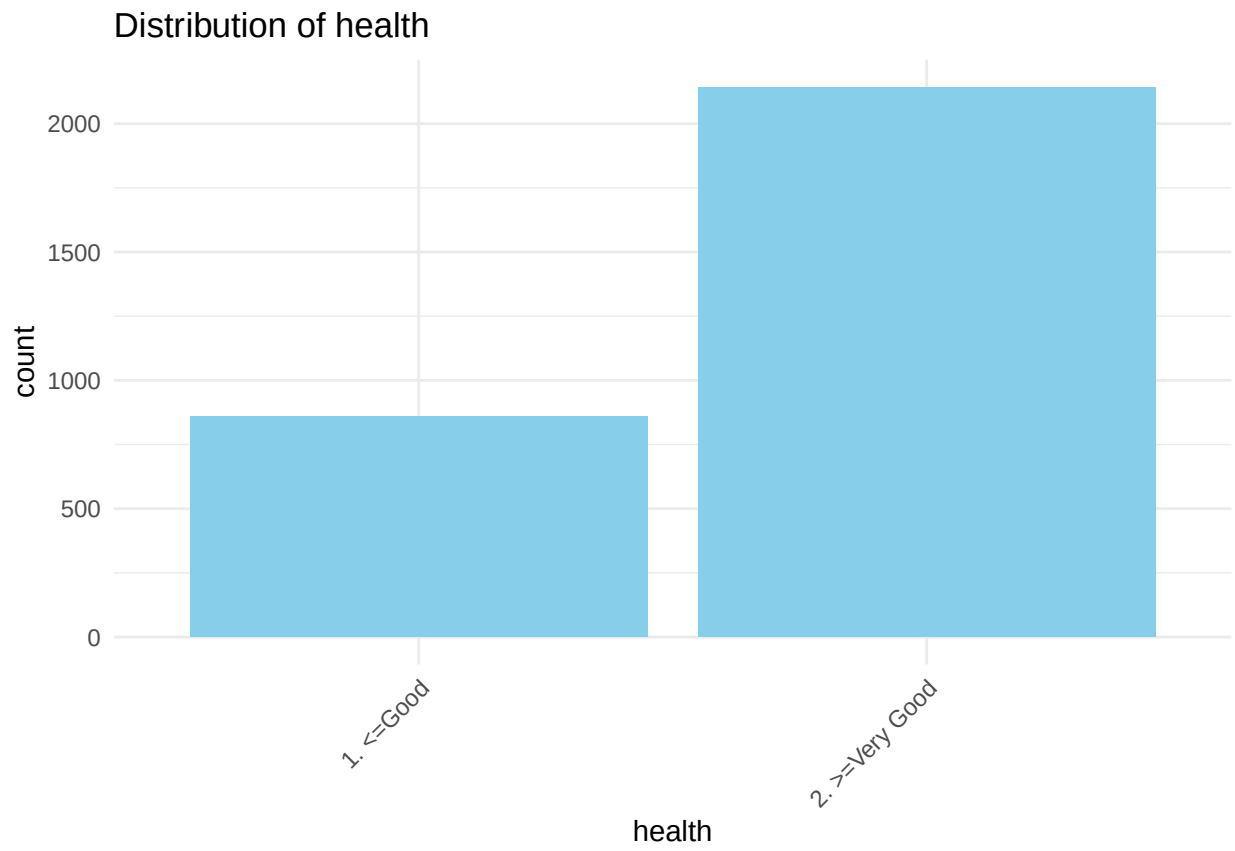
```
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.  
## i Please use tidy evaluation idioms with 'aes()'.  
## i See also 'vignette("ggplot2-in-packages")' for more information.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

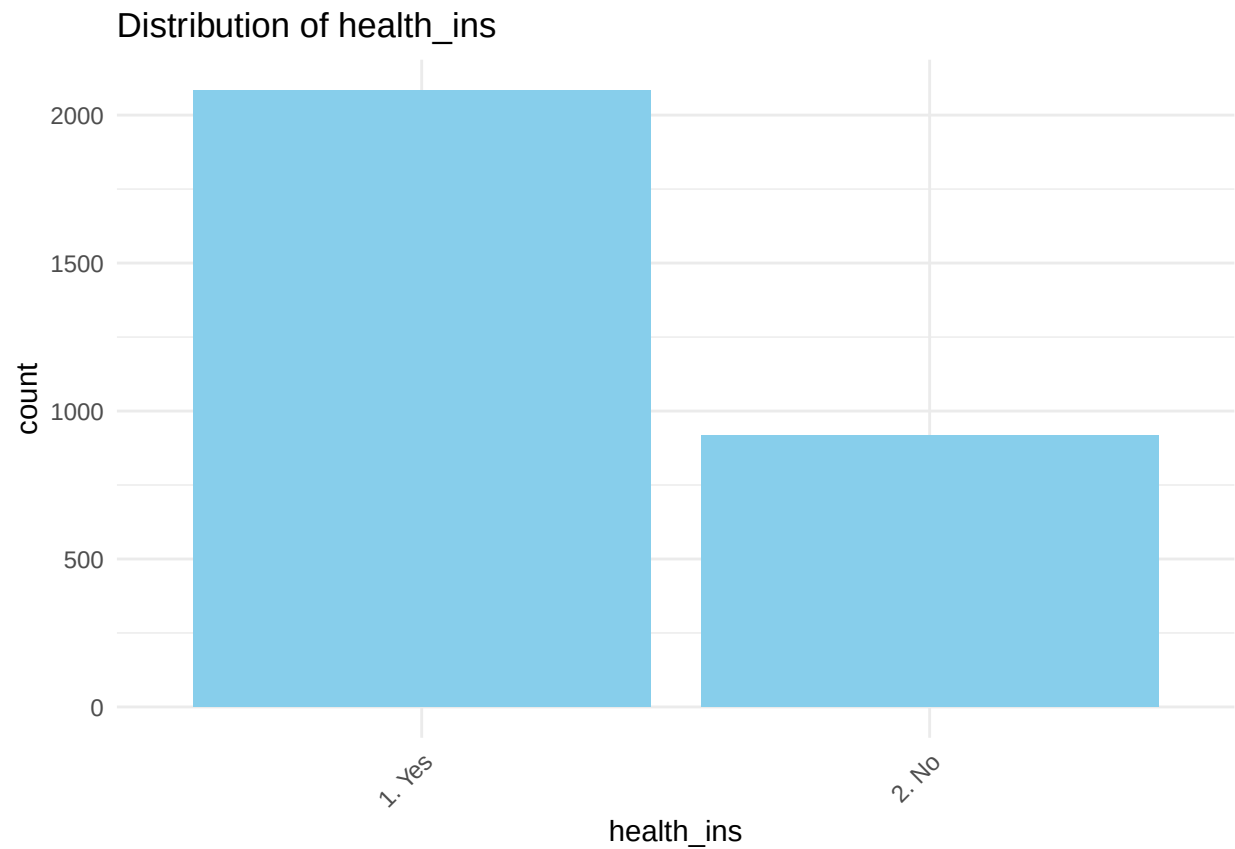








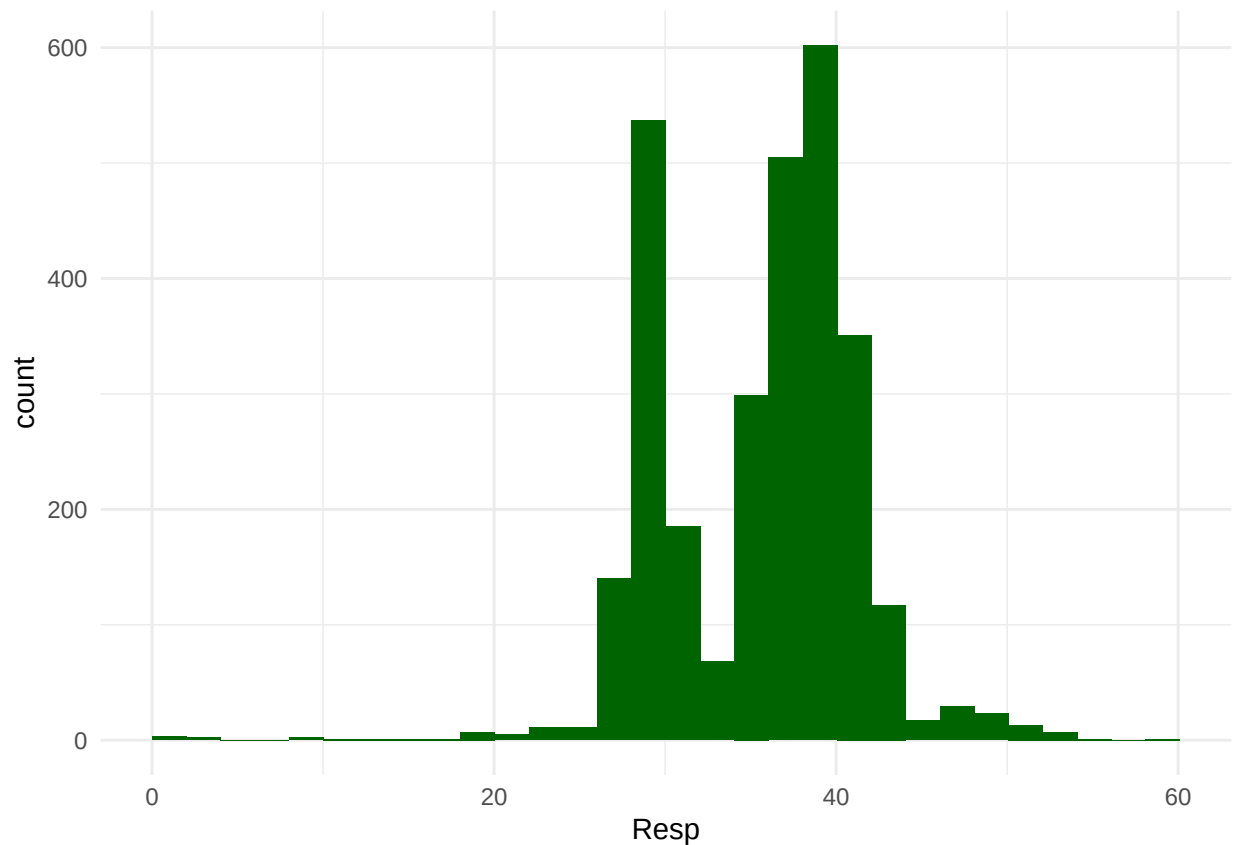




Distribution of Resp

```
ggplot(Wage, aes(Resp)) +  
  geom_histogram(bins = 30, fill = "darkgreen") +  
  theme_minimal()
```

```
## Warning: Removed 60 rows containing non-finite outside the scale range  
## ('stat_bin()').
```

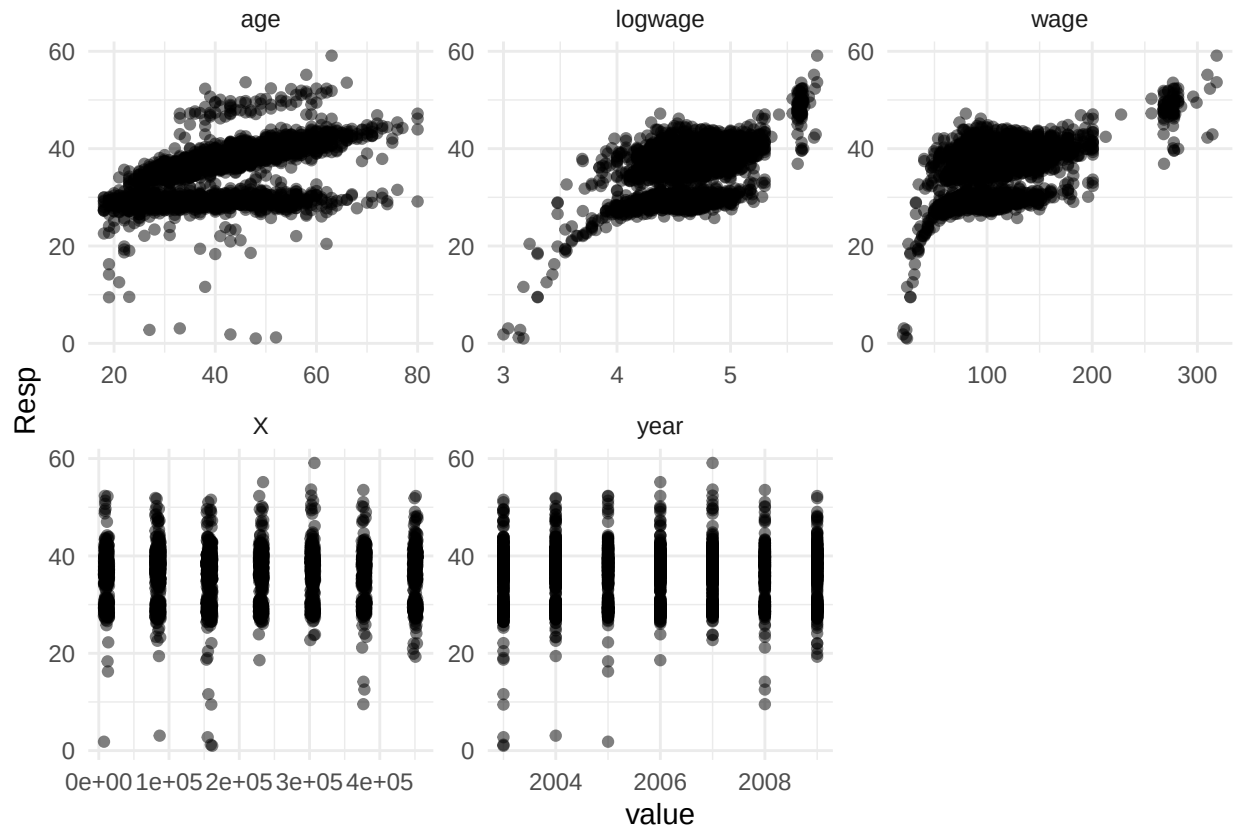


Scatterplots of numeric predictors vs Resp

```
numeric_vars <- Wage %>% select(where(is.numeric))
```

```
numeric_vars %>%  
  pivot_longer(cols = -Resp) %>%  
  ggplot(aes(x = value, y = Resp)) +  
  geom_point(alpha = 0.5) +  
  facet_wrap(~name, scales = "free") +  
  theme_minimal()
```

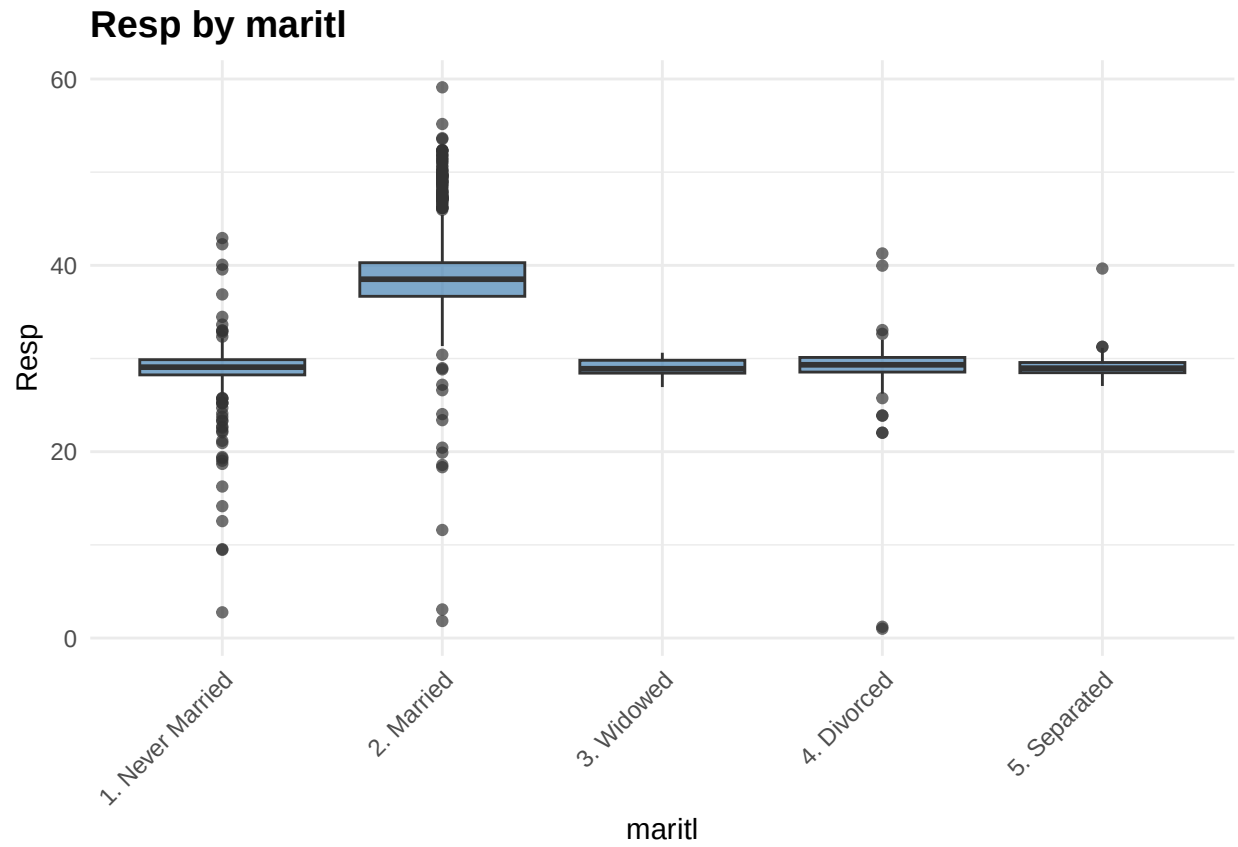
```
## Warning: Removed 300 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



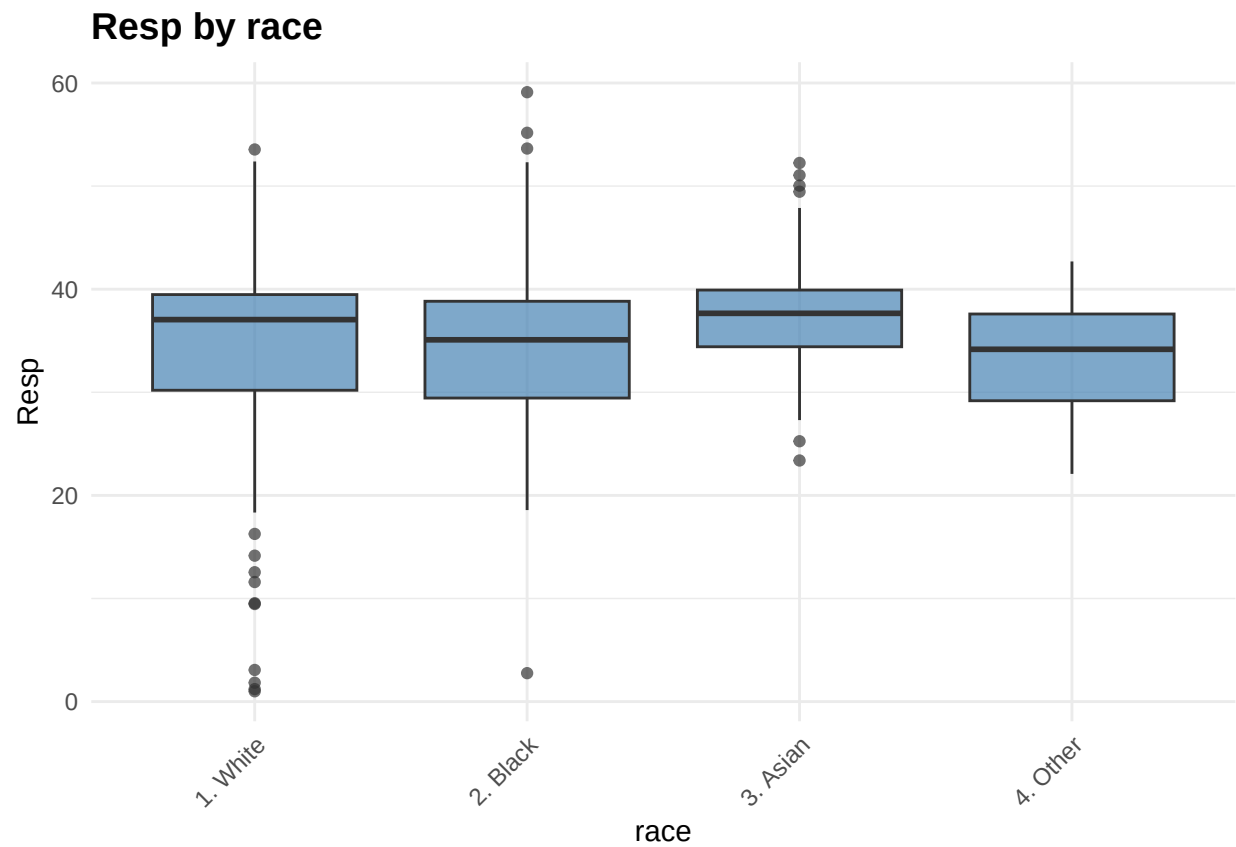
Boxplots of categorical predictors vs Resp

```
for (v in cat_vars) {
  print(
    Wage %>%
      ggplot(aes(x = .data[[v]], y = Resp)) +
      geom_boxplot(fill = "steelblue", alpha = 0.7) +
      theme_minimal() +
      ggtitle(paste("Resp by", v)) +
      theme(
        axis.text.x = element_text(angle = 45, hjust = 1),
        plot.title = element_text(size = 14, face = "bold")
      )
  )
}
```

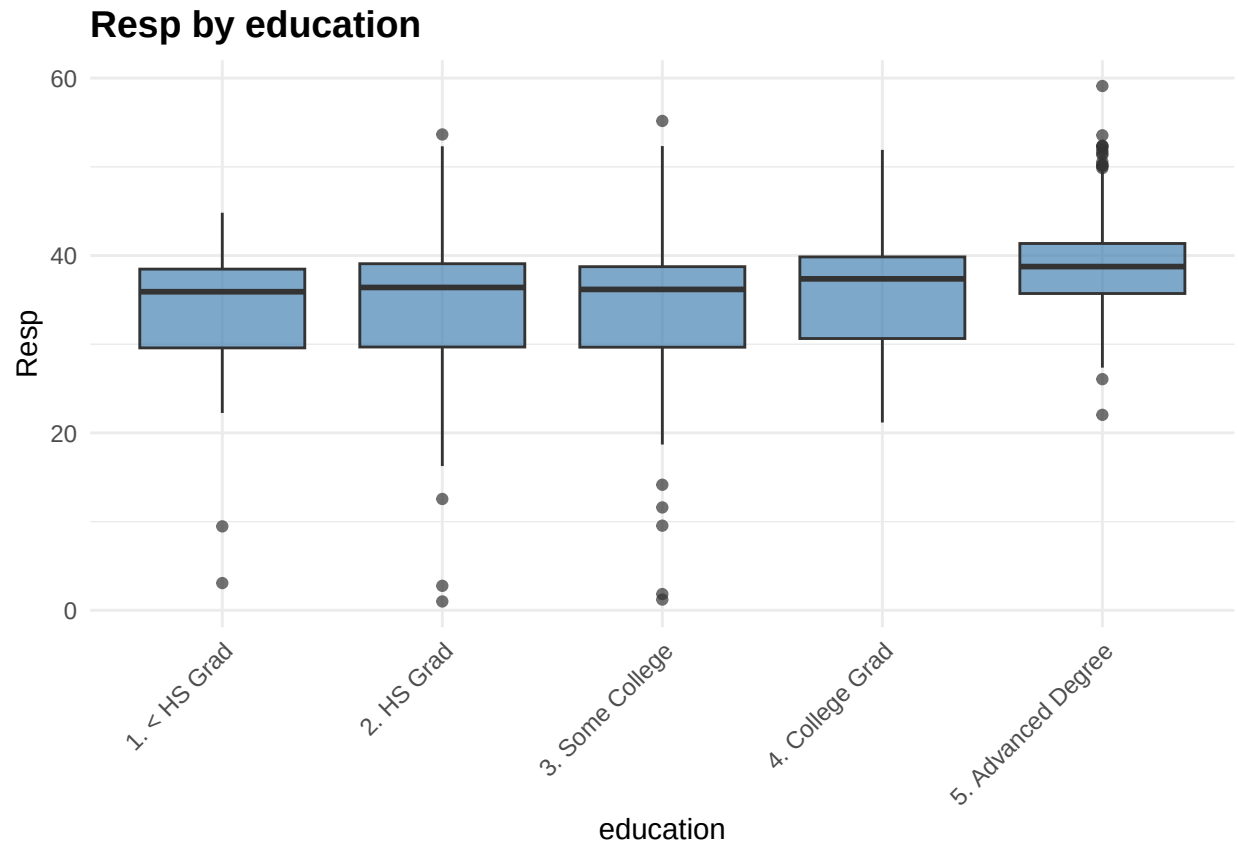
```
## Warning: Removed 60 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



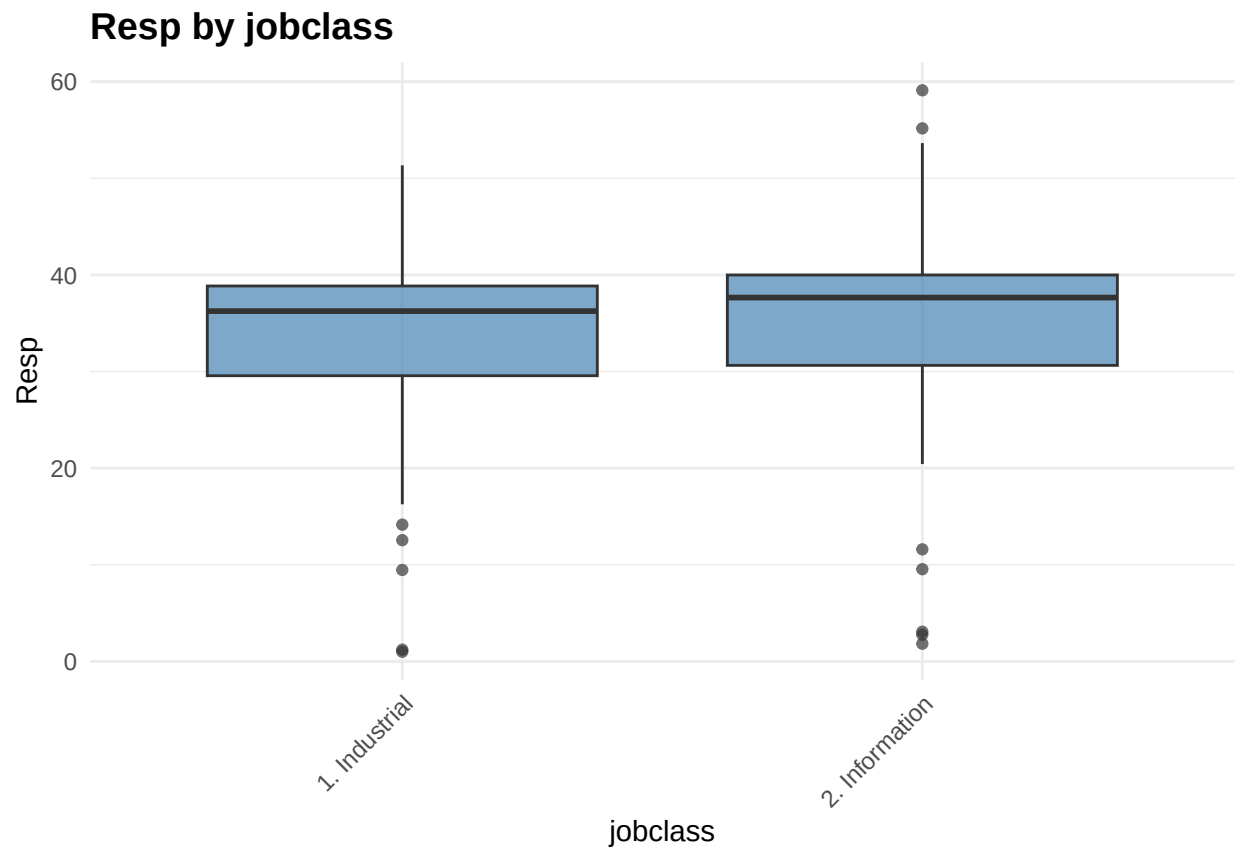
```
## Warning: Removed 60 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```



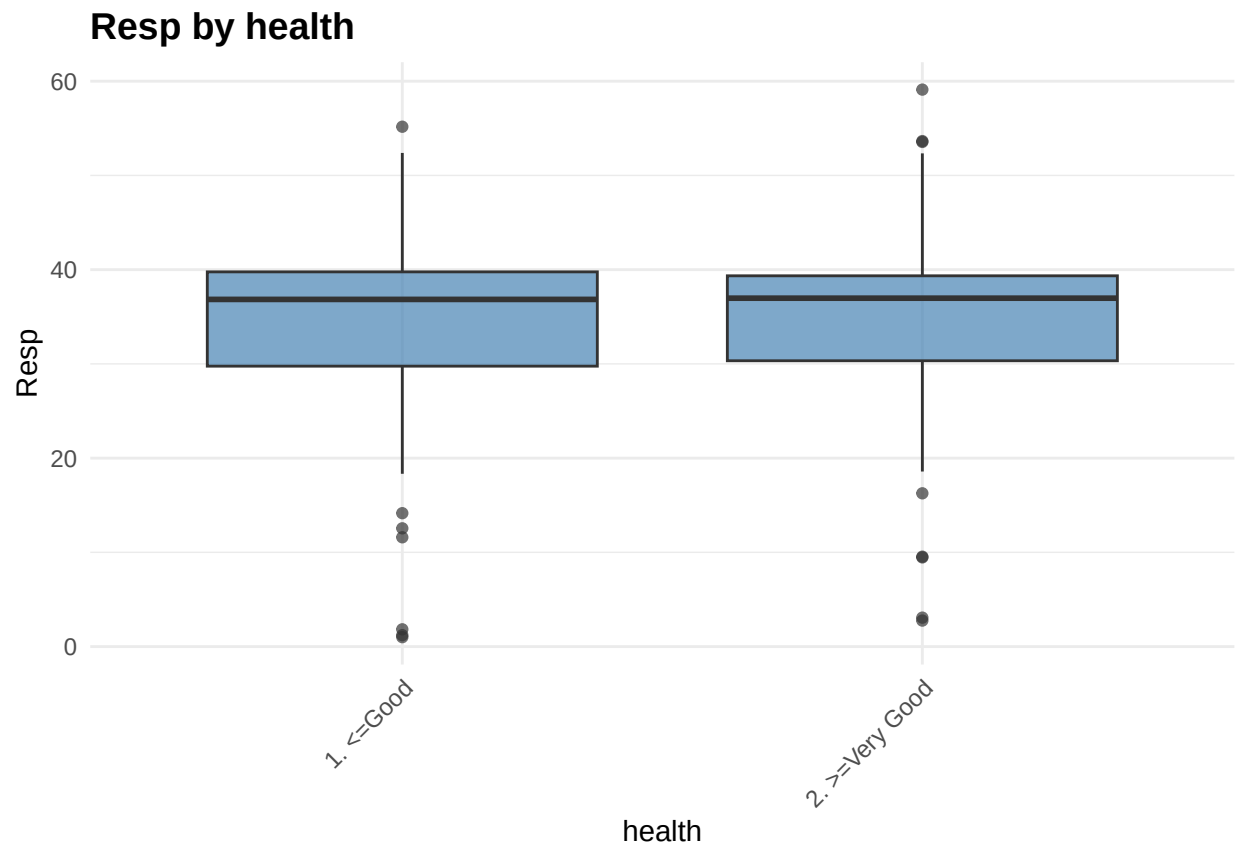
```
## Warning: Removed 60 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



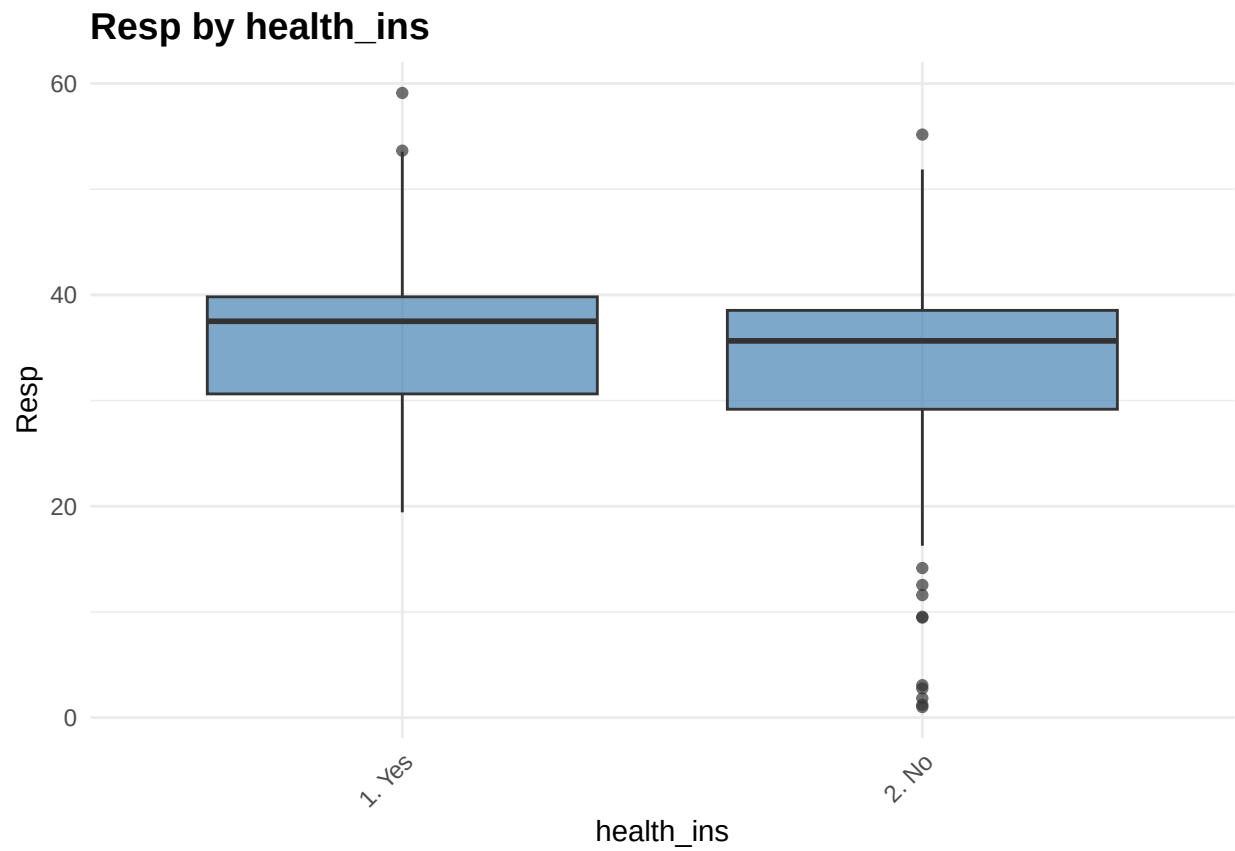
```
## Warning: Removed 60 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
## Warning: Removed 60 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

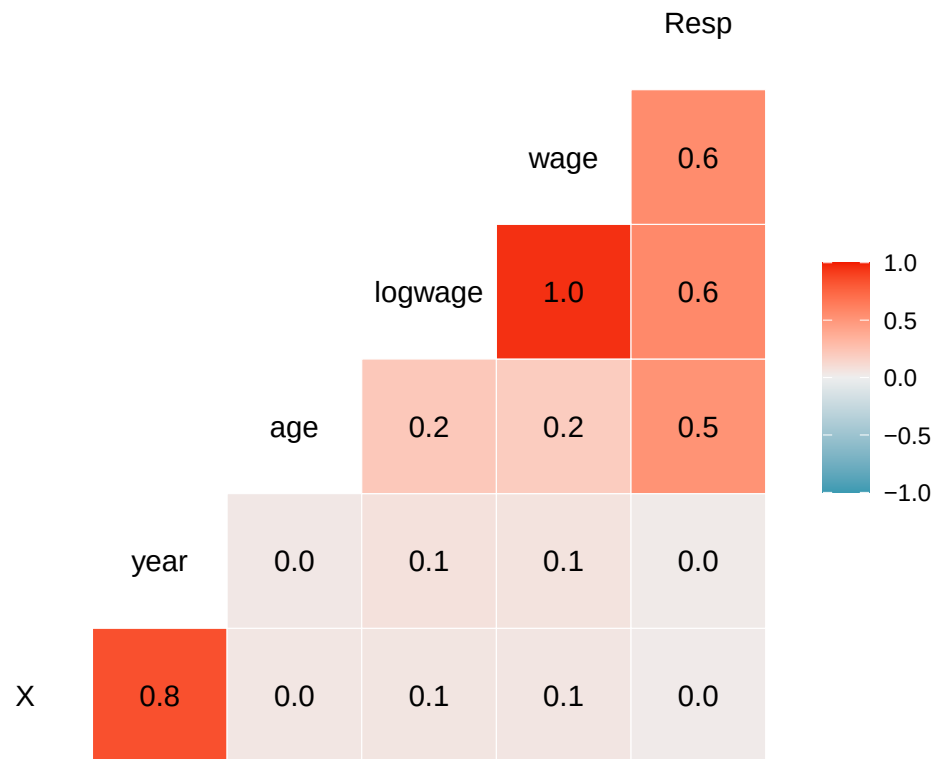


```
## Warning: Removed 60 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

Correlation matrix

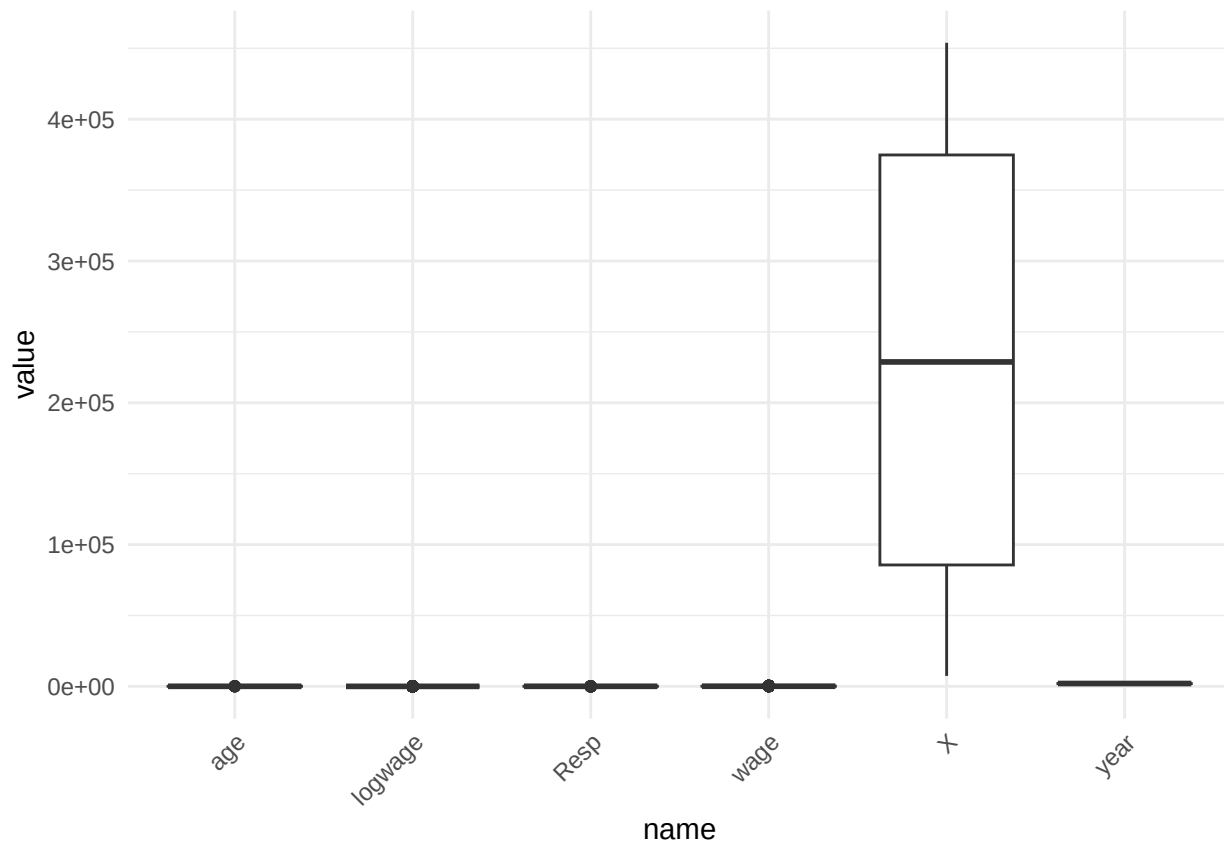
```
ggcorr(numeric_vars, label = TRUE)
```



Outliers

```
numeric_vars %>%
  pivot_longer(everything()) %>%
  ggplot(aes(x = name, y = value)) +
  geom_boxplot() +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: Removed 60 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



Preprocessing

```
Wage_clean <- Wage %>% filter(!is.na(Resp))
```

```
train_index <- createDataPartition(Wage_clean$Resp, p = 0.80, list = FALSE)
```

```
train <- Wage_clean[train_index, ]
```

```
test <- Wage_clean[-train_index, ]
```

```
dim(train)
```

```
## [1] 2352 12
```

```
dim(test)
```

```
## [1] 588 12
```

```
# Final dataset ready for modeling
```

```
head(train)
```

```
##      X year age      maritl    race      education      jobclass
## 1 231655 2006  18 1. Never Married 1. White    1. < HS Grad  1. Industrial
## 3 161300 2003  45      2. Married 1. White  3. Some College  1. Industrial
## 4 155159 2003  43      2. Married 3. Asian  4. College Grad  2. Information
## 5  11443 2005  50      4. Divorced 1. White    2. HS Grad  2. Information
```

```
## 6 376662 2008 54      2. Married 1. White 4. College Grad 2. Information
## 7 450601 2009 44      2. Married 4. Other 3. Some College 1. Industrial
##      health health_ins logwage      wage      Resp
## 1      1. <=Good      2. No 4.318063  75.04315 28.024
## 3      1. <=Good      1. Yes 4.875061 130.98218 36.118
## 4 2. >=Very Good      1. Yes 5.041393 154.68529 38.678
## 5      1. <=Good      1. Yes 4.318063  75.04315 29.526
## 6 2. >=Very Good      1. Yes 4.845098 127.11574 41.816
## 7 2. >=Very Good      1. Yes 5.133021 169.52854 39.080
```

Modeling

Baseline MLR

```
x_train <- train %>% select(-Resp)
y_train <- train$Resp
x_test  <- test  %>% select(-Resp)
y_test  <- test$Resp

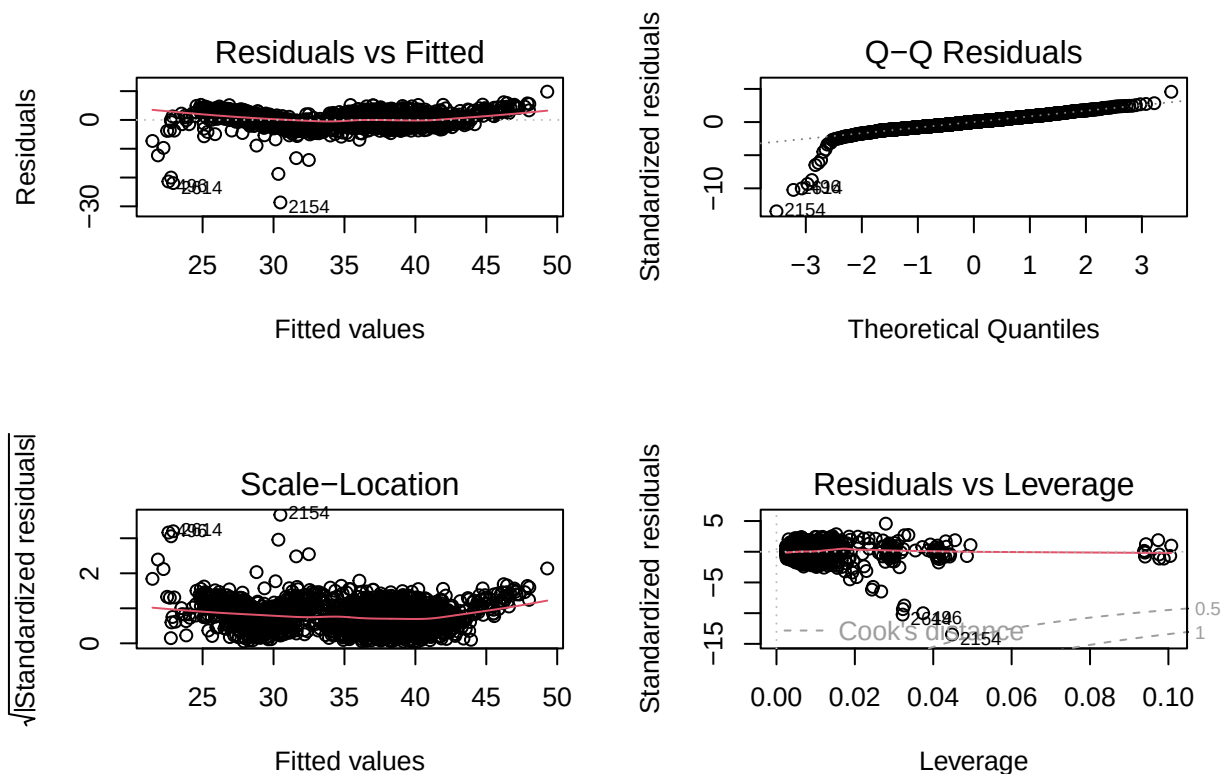
# Fit full model with all predictors
lm_fit <- lm(Resp ~ ., data = train)

summary(lm_fit)           # coefficients, significance
```

```
##
## Call:
## lm(formula = Resp ~ ., data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -28.6519  -1.2042   0.0729   1.2749   9.8073
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -5.923e+00  8.270e+01  -0.072  0.942910
## X             -8.750e-07  5.723e-07  -1.529  0.126427
## year           6.942e-03  4.133e-02   0.168  0.866632
## age            1.361e-01  4.524e-03  30.079 < 2e-16 ***
## maritl2. Married  6.794e+00  1.262e-01  53.845 < 2e-16 ***
## maritl3. Widowed -2.549e+00  6.695e-01  -3.808  0.000144 ***
## maritl4. Divorced -2.329e+00  2.088e-01 -11.157 < 2e-16 ***
## maritl5. Separated -1.727e+00  3.514e-01  -4.915  9.49e-07 ***
## race2. Black      1.158e-01  1.568e-01   0.739  0.460183
## race3. Asian      1.108e-01  1.874e-01   0.591  0.554361
## race4. Other     -1.147e-01  4.168e-01  -0.275  0.783174
## education2. HS Grad -5.117e-01  1.728e-01  -2.962  0.003091 **
## education3. Some College -1.228e+00  1.851e-01  -6.631  4.14e-11 ***
## education4. College Grad -1.380e+00  1.917e-01  -7.201  8.05e-13 ***
## education5. Advanced Degree -1.424e+00  2.175e-01  -6.548  7.13e-11 ***
## jobclass2. Information  3.652e-01  9.704e-02   3.763  0.000172 ***
## health2. >=Very Good  3.291e-03  1.029e-01   0.032  0.974497
## health_ins2. No     2.220e-01  1.077e-01   2.061  0.039451 *
## logwage         3.329e+00  4.351e-01   7.653  2.86e-14 ***
## wage           2.495e-02  3.582e-03   6.967  4.20e-12 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.177 on 2332 degrees of freedom
## Multiple R-squared:  0.8507, Adjusted R-squared:  0.8495
## F-statistic: 699.2 on 19 and 2332 DF,  p-value: < 2.2e-16
```

```
par(mfrow = c(2, 2))
plot(lm_fit)           # residual diagnostics
```



```
par(mfrow = c(1, 1))

# Test-set performance
lm_pred <- predict(lm_fit, newdata = test)
lm_mse  <- mean((y_test - lm_pred)^2)
lm_mse
```

```
## [1] 5.465108
```

Ridge and LASSO

```
library(glmnet)
```

```
## Loading required package: Matrix
```

```
##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

## Loaded glmnet 4.1-10

# Model matrix (handles factors with dummies)
x_train_mat <- model.matrix(Resp ~ ., data = train)[, -1]
x_test_mat  <- model.matrix(Resp ~ ., data = test)[, -1]

# Ridge (alpha = 0)
set.seed(4620)
ridge_cv <- cv.glmnet(x_train_mat, y_train, alpha = 0)
ridge_best_lambda <- ridge_cv$lambda.min

ridge_pred <- predict(ridge_cv, s = ridge_best_lambda, newx = x_test_mat)
ridge_mse  <- mean((y_test - ridge_pred)^2)
ridge_mse

## [1] 5.480126

# LASSO (alpha = 1)
set.seed(4620)
lasso_cv <- cv.glmnet(x_train_mat, y_train, alpha = 1)
lasso_best_lambda <- lasso_cv$lambda.min

lasso_pred <- predict(lasso_cv, s = lasso_best_lambda, newx = x_test_mat)
lasso_mse  <- mean((y_test - lasso_pred)^2)
lasso_mse

## [1] 5.448802

# Optional: see which variables LASSO keeps
lasso_coefs <- coef(lasso_cv, s = lasso_best_lambda)
lasso_coefs

## 20 x 1 sparse Matrix of class "dgCMatrix"
##                                s=0.006698545
## (Intercept)                   8.105734e+00
## X                             -7.315083e-07
## year                          .
## age                           1.354810e-01
## maritl2. Married                6.805370e+00
## maritl3. Widowed               -2.435916e+00
## maritl4. Divorced              -2.298095e+00
## maritl5. Separated             -1.624049e+00
## race2. Black                   9.707779e-02
## race3. Asian                   8.005275e-02
```

```
## race4. Other -2.561664e-02
## education2. HS Grad -3.514000e-01
## education3. Some College -1.055474e+00
## education4. College Grad -1.195132e+00
## education5. Advanced Degree -1.216862e+00
## jobclass2. Information 3.404970e-01
## health2. >=Very Good .
## health_ins2. No 2.044151e-01
## logwage 3.275532e+00
## wage 2.485697e-02
```

Regression tree and random forest

```
library(rpart)
```

```
## Warning: package 'rpart' was built under R version 4.5.2
```

```
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.5.2
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
```

```
## Attaching package: 'randomForest'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
## combine
```

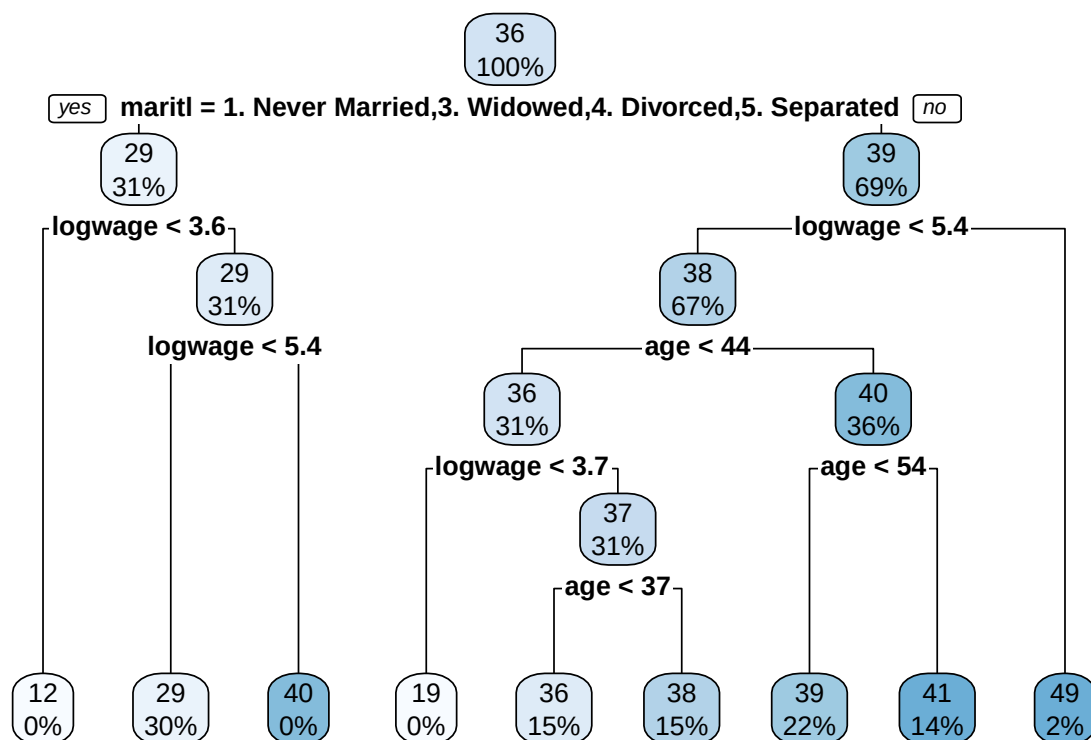
```
## The following object is masked from 'package:ggplot2':
```

```
##
```

```
## margin
```

```
# Regression tree
```

```
tree_fit <- rpart(Resp ~ ., data = train, method = "anova")
rpart.plot(tree_fit)
```



```
tree_pred <- predict(tree_fit, newdata = test)
tree_mse <- mean((y_test - tree_pred)^2)
tree_mse
```

```
## [1] 3.364042
```

```
# Random forest
set.seed(4620)
rf_fit <- randomForest(Resp ~ ., data = train,
                       ntree = 500,
                       importance = TRUE)

rf_pred <- predict(rf_fit, newdata = test)
rf_mse <- mean((y_test - rf_pred)^2)
rf_mse
```

```
## [1] 1.615084
```

```
importance(rf_fit)
```

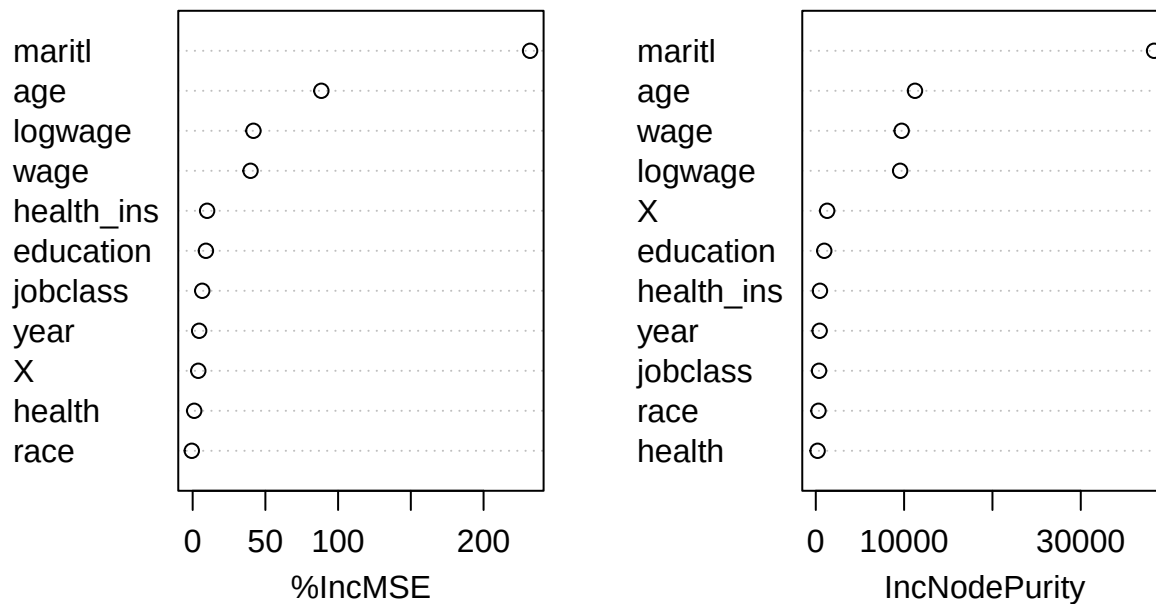
```
##           %IncMSE IncNodePurity
## X           3.8736511      1289.4212
## year         4.4728305       436.5168
## age          88.4872608     11230.1852
```



```
## marital    231.9735999    38297.3211
## race       -0.5997994     309.1458
## education   9.1170847     959.0769
## jobclass    6.5860689     372.5826
## health      1.0684171     206.9978
## health_ins   9.9487997     471.7949
## logwage     41.7810741    9549.9230
## wage        39.7901788    9725.1029
```

```
varImpPlot(rf_fit)
```

rf_fit



GAMs

```
library(mgcv)
```

```
## Warning: package 'mgcv' was built under R version 4.5.2
```

```
## Loading required package: nlme
```

```
##
```

```
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
## collapse
```

```
## This is mgcv 1.9-4. For overview type '?mgcv'.
```

```
# Quick check of unique values
```

```
sapply(train[, c("age", "year", "logwage")], function(x) length(unique(x)))
```

```
##      age      year logwage
##      61         7      447
```

```
set.seed(4620)
```

```
# 1) Simple GAM with smooth age only (small k)
```

```
gam1 <- gam(Resp ~ s(age, k = 5), data = train)
```

```
summary(gam1)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## Resp ~ s(age, k = 5)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 35.65087    0.09736   366.2   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(age) 3.767   3.967 244.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.292   Deviance explained = 29.3%
## GCV = 22.341   Scale est. = 22.296      n = 2352
```

```
gam1_pred <- predict(gam1, newdata = test)
gam1_mse <- mean((test$Resp - gam1_pred)^2)
gam1_mse
```

```
## [1] 23.06986
```

```
# 2) GAM with smooths for age and year, plus factors
```

```
gam2 <- gam(Resp ~ s(age, k = 5) + s(year, k = 5) +
            maritl + race + education + jobclass + health + health_ins,
            data = train)
```

```
summary(gam2)
```

```
##
## Family: gaussian
## Link function: identity
```

```
##
## Formula:
## Resp ~ s(age, k = 5) + s(year, k = 5) + maritl + race + education +
##      jobclass + health + health_ins
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    29.79676    0.25893 115.075 < 2e-16 ***
## maritl2. Married      7.99409    0.16552  48.296 < 2e-16 ***
## maritl3. Widowed     -2.30582    0.85433  -2.699  0.00701 **
## maritl4. Divorced     -1.91932    0.27122 -7.077 1.95e-12 ***
## maritl5. Separated    -0.83724    0.45036  -1.859  0.06315 .
## race2. Black         -0.15985    0.19994  -0.800  0.42408
## race3. Asian          0.05303    0.23909   0.222  0.82450
## race4. Other         -0.60194    0.53103  -1.134  0.25711
## education2. HS Grad   -0.03308    0.21967  -0.151  0.88030
## education3. Some College -0.15844    0.23303  -0.680  0.49664
## education4. College Grad  0.40922    0.23744   1.723  0.08494 .
## education5. Advanced Degree 1.39093    0.26197   5.310 1.20e-07 ***
## jobclass2. Information  0.57802    0.12360   4.677 3.08e-06 ***
## health2. >=Very Good   0.35830    0.13075   2.740  0.00618 **
## health_ins2. No       -0.88566    0.13134  -6.743 1.95e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(age)      3.178  3.651 195.672 <2e-16 ***
## s(year)     1.957  2.400   1.803   0.193
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.755   Deviance explained = 75.7%
## GCV = 7.7668   Scale est. = 7.7003    n = 2352

gam2_pred <- predict(gam2, newdata = test)
gam2_mse <- mean((test$Resp - gam2_pred)^2)
gam2_mse

## [1] 8.804283

# 3) GAM with smooths for age and logwage
gam3 <- gam(Resp ~ s(age, k = 5) + s(logwage, k = 5) +
            maritl + race + education + jobclass + health + health_ins,
            data = train)
summary(gam3)

##
## Family: gaussian
## Link function: identity
##
## Formula:
## Resp ~ s(age, k = 5) + s(logwage, k = 5) + maritl + race + education +
```

```
##      jobclass + health + health_ins
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    30.24391    0.13455 224.774 < 2e-16 ***
## maritl2. Married      7.73353    0.08475  91.252 < 2e-16 ***
## maritl3. Widowed     -2.11162    0.42755  -4.939 8.41e-07 ***
## maritl4. Divorced     -1.53405    0.13604 -11.276 < 2e-16 ***
## maritl5. Separated    -1.28522    0.22571  -5.694 1.40e-08 ***
## race2. Black         -0.07220    0.10021  -0.720   0.471
## race3. Asian          0.10179    0.11973   0.850   0.395
## race4. Other         -0.39467    0.26621  -1.483   0.138
## education2. HS Grad    0.10852    0.11104   0.977   0.329
## education3. Some College 0.02930    0.12051   0.243   0.808
## education4. College Grad 0.01165    0.12487   0.093   0.926
## education5. Advanced Degree -0.13487    0.14089  -0.957   0.339
## jobclass2. Information  0.44541    0.06193   7.192 8.54e-13 ***
## health2. >=Very Good   -0.00275    0.06567  -0.042   0.967
## health_ins2. No       -0.09582    0.06902  -1.388   0.165
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(age)        3.892  3.993  643.7 <2e-16 ***
## s(logwage)    3.994  4.000 1748.7 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.939   Deviance explained = 93.9%
## GCV = 1.9473   Scale est. = 1.9284    n = 2352
```

```
gam3_pred <- predict(gam3, newdata = test)
gam3_mse <- mean((test$Resp - gam3_pred)^2)
gam3_mse
```

```
## [1] 1.957593
```

```
# 4) GAM with smooths for all three numeric predictors (still small k)
gam4 <- gam(Resp ~ s(age, k = 5) + s(year, k = 5) + s(logwage, k = 5) +
             maritl + race + education + jobclass + health + health_ins,
             data = train)
summary(gam4)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## Resp ~ s(age, k = 5) + s(year, k = 5) + s(logwage, k = 5) + maritl +
##      race + education + jobclass + health + health_ins
##
## Parametric coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      30.251123   0.134559  224.817 < 2e-16 ***
## maritl2. Married    7.727978   0.084772   91.162 < 2e-16 ***
## maritl3. Widowed   -2.134698   0.427572  -4.993 6.40e-07 ***
## maritl4. Divorced  -1.539967   0.136028 -11.321 < 2e-16 ***
## maritl5. Separated -1.287770   0.225616  -5.708 1.29e-08 ***
## race2. Black       -0.066664   0.100220  -0.665   0.506
## race3. Asian        0.108446   0.119746   0.906   0.365
## race4. Other       -0.393616   0.266092  -1.479   0.139
## education2. HS Grad  0.106677   0.110999   0.961   0.337
## education3. Some College 0.022836   0.120521   0.189   0.850
## education4. College Grad 0.006427   0.124853   0.051   0.959
## education5. Advanced Degree -0.137787   0.140841  -0.978   0.328
## jobclass2. Information 0.441993   0.061933   7.137 1.27e-12 ***
## health2. >=Very Good -0.002748   0.065642  -0.042   0.967
## health_ins2. No     -0.090936   0.069052  -1.317   0.188
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##           edf Ref.df      F p-value
## s(age)      3.890  3.993 644.746 <2e-16 ***
## s(year)      1.000  1.000   3.004  0.0832 .
## s(logwage)   3.994  4.000 1750.616 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.939   Deviance explained = 93.9%
## GCV = 1.9465   Scale est. = 1.9267     n = 2352
```

```
gam4_pred <- predict(gam4, newdata = test)
gam4_mse <- mean((test$Resp - gam4_pred)^2)
gam4_mse
```

```
## [1] 1.950388
```

```
gam_results <- tibble(
  Model = c("Linear regression", "GAM age", "GAM age+year+factors",
            "GAM age+logwage+factors", "GAM age+year+logwage+factors"),
  Test_MSE = c(lm_mse, gam1_mse, gam2_mse, gam3_mse, gam4_mse)
) %>%
  arrange(Test_MSE)

gam_results
```

```
## # A tibble: 5 x 2
##   Model                Test_MSE
##   <chr>                <dbl>
## 1 GAM age+year+logwage+factors    1.95
## 2 GAM age+logwage+factors        1.96
## 3 Linear regression             5.47
## 4 GAM age+year+factors           8.80
## 5 GAM age                      23.1
```

```

set.seed(4620)

library(mgcv)
library(caret)

# candidate GAM formulas (all entries are formulas only)
gam_forms <- list(
  gam1 = Resp ~ s(age, k = 5),

  gam2 = Resp ~ s(age, k = 5) + s(year, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  gam3 = Resp ~ s(age, k = 5) + s(logwage, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  gam4 = Resp ~ s(age, k = 5) + s(year, k = 5) + s(logwage, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  gam5 = Resp ~ s(logwage, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  gam6 = Resp ~ s(age, k = 5) + s(year, k = 5) + s(logwage, k = 5) + s(wage, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  # interaction models among numeric predictors
  gam7 = Resp ~ s(age, k = 5) + s(year, k = 5) + ti(age, year, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  gam8 = Resp ~ s(age, k = 5) + s(logwage, k = 5) + ti(age, logwage, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  gam9 = Resp ~ s(age, k = 5) + s(year, k = 5) + s(logwage, k = 5) + s(wage, k = 5) +
    ti(age, logwage, k = 5) + ti(age, year, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  # all pairwise numeric interactions
  gam10 = Resp ~
    s(age, k = 5) + s(year, k = 5) + s(logwage, k = 5) + s(wage, k = 5) +
    ti(age, year, k = 5) +
    ti(age, logwage, k = 5) +
    ti(age, wage, k = 5) +
    ti(year, logwage, k = 5) +
    ti(year, wage, k = 5) +
    ti(logwage, wage, k = 5) +
    maritl + race + education + jobclass + health + health_ins,

  # categorical interactions
  gam11 = Resp ~
    s(age, k = 5) + s(year, k = 5) + s(logwage, k = 5) +
    maritl * education +
    education * jobclass +
    race + health + health_ins,

```

```

# varying-by-maritl smooths for age and logwage
gam12 = Resp ~
  s(logwage, by = maritl, k = 5) +
  s(age,      by = maritl, k = 5) +
  maritl + education + race + jobclass + health + health_ins
)

K <- 5 # 5-fold CV
folds <- createFolds(train$Resp, k = K, list = TRUE, returnTrain = FALSE)

cv_results <- tibble(Model = character(), CV_MSE = numeric())

for (m in names(gam_forms)) {
  form <- gam_forms[[m]]
  mse_vec <- numeric(K)

  for (i in seq_along(folds)) {
    val_idx <- folds[[i]]
    train_cv <- train[-val_idx, ]
    val_cv <- train[val_idx, ]

    fit_cv <- gam(form, data = train_cv)
    pred_cv <- predict(fit_cv, newdata = val_cv)
    mse_vec[i] <- mean((val_cv$Resp - pred_cv)^2)
  }

  cv_results <- cv_results %>%
    add_row(Model = m, CV_MSE = mean(mse_vec))
}

cv_results <- cv_results %>% arrange(CV_MSE)
cv_results

```

```

## # A tibble: 12 x 2
##   Model CV_MSE
##   <chr> <dbl>
## 1 gam12  1.24
## 2 gam6   1.81
## 3 gam9   1.85
## 4 gam10  1.85
## 5 gam4   1.98
## 6 gam3   1.98
## 7 gam8   2.00
## 8 gam11  2.00
## 9 gam5   4.10
## 10 gam2  7.74
## 11 gam7  7.79
## 12 gam1 22.3

```

```

best_model_name <- cv_results$Model[1]
best_form <- gam_forms[[best_model_name]]
best_model_name

```

```
## [1] "gam12"
```

```
best_form
```

```
## Resp ~ s(logwage, by = maritl, k = 5) + s(age, by = maritl, k = 5) +  
##      maritl + education + race + jobclass + health + health_ins
```

```
# Refit on all training data
```

```
gam_best <- gam(best_form, data = train)
```

```
# Test-set performance
```

```
gam_best_pred <- predict(gam_best, newdata = test)
```

```
gam_best_mse <- mean((test$Resp - gam_best_pred)^2)
```

```
gam_best_mse
```

```
## [1] 1.182728
```